

Name: Prabhakant Singh	Mob. No.: 6388011997	Location: Kanpur Dehat	Village: Rampur
Previous Crop: Moong	Farm Size: 1.00 Bigha	Farm Name: Prabhakant Singh Farm	Next Crop: Rice
Date of Collection: 2024-07-09		Sample Code: S0002515	Treatment Type: Organic

SOIL ANALYSIS PARAMETERS

The table explains each nutrient, its role. Important to understand your soil better.

S.No.	Parameter	Ideal Range	Actual Value	Status
1	pH	6.5 – 7.5	7.2	
2	EC (dS/m)	1 – 3	0.8	
3	OC (%)	0.5 – 0.75	0.4	
4	Nitrogen	280 – 560 kg/ha	245 kg/ha	
5	Phosphorus (P,O...)	28 – 56 kg/ha	42 kg/ha	
6	Potassium (K,O)	110 – 280 kg/ha	156 kg/ha	
7	Sulfur (S)	10 – 20 mg/kg	15 mg/kg	
8	Zinc (Zn)	0.6 – 1.2 mg/kg	0.5 mg/kg	
9	Iron (Fe)	4.5 – 15 mg/kg	8.2 mg/kg	
10	Manganese (Mn)	1.0 – 20 mg/kg	3.5 mg/kg	
11	Copper (Cu)	0.2 – 0.5 mg/kg	0.3 mg/kg	
12	Boron (B)	0.5 – 1.0 mg/kg	0.7 mg/kg	
13	1OC (%)	0.5 – 0.75	0.4	

DERIVED PARAMETERS

14	C:N Ratio	10:1–12:1	13:1	
15	CEC Tendency	Medium to High	-	
16	Ca:Mg Ratio	5:1–7:1	H 5:1	
17	K:Mg Ratio	0.3 to 0.7	H 0.57	

Low

Medium

High

SECTION 2: DETAILED NUTRIENT INSIGHTS

This page explains each nutrient and its role. Important to understand your soil better.

Parameter	Fertility rating	What it means?	Function in Crop Growth
pH	LowMediumHigh	Acidity/alkalinity of soil	Affects nutrient availability and microbial activity
EC (dS/m)	LowMediumHigh	Salt level in soil	Influences root water uptake and nutrient solubility
OC (%)	LowMediumHigh	Soil organic matter	Enhances microbial life, structure, and water retention
Nitrogen	LowMediumHigh	Growth nutrient	Promotes leafy growth and chlorophyll formation
Phosphorus	LowMediumHigh	Energy transfer	Supports root development and flowering
Potassium	LowMediumHigh	Fruit quality regulator	Helps in water balance, disease resistance, fruit size
Sulphur	LowMediumHigh	Protein synthesis	Improves enzyme function, oil synthesis, and taste
Magnesium	LowMediumHigh	Core of chlorophyll	Essential for photosynthesis and enzyme activation
Calcium	LowMediumHigh	Cell strength	Builds strong cell walls, prevents blossom-end rot
Zinc	LowMediumHigh	Growth hormone support	Needed for enzyme activation, leaf expansion, rooting
Iron	LowMediumHigh	Chlorophyll synthesis	Supports green leaf formation and respiration
Copper	LowMediumHigh	Enzyme activator	Influences pollen viability and disease resistance
Boron	LowMediumHigh	Pollination & fruit set	Aids flowering, fruit set, sugar transport, and seed formation

Status of Other derived parameters

Parameter	Fertility rating	What it means?	Function in Crop Growth
C:N Ratio	LowMediumHigh	Ratio of soil Organic carbon to available nitrogen	Indicates quality of organic matter; affects microbial activity and nitrogen availability
CEC Tendency	LowMediumHigh	Soil's capacity to hold and exchange nutrients (based on OC, Ca, Mg, K, pH)	Influences root water uptake and nutrient solubility
Ca:Mg Ratio	LowMediumHigh	Balance between calcium and magnesium levels in soil	Influences soil structure, aeration, water movement, and root penetration
K:Mg Ratio	LowMediumHigh	Balance between potassium and magnesium availability	Controls competition for uptake; affects fruit quality, stress tolerance, and leaf strength

SECTION 5: CROP-SPECIFIC TIPS (POTATO)

Expert tips for potato cultivation. Follow these practices for better results.

Stage	Practice	Why It Matters
Before Planting	Soil Testing	Identifies nutrient deficiencies; potatoes need well-balanced nutrition for tuber formation.
Before Planting	Seed Treatment (Fungicide + Insecticide + Biofertilizer)	Protects against seed-borne diseases and soil-borne pests; enhances early root health.
Before Planting	Field Prep (Deep Ploughing + Fine Tilth + Ridges)	Ensures aeration, drainage, and uniform tuber development; prevents waterlogging.
Before Planting	Apply FYM/Compost + Gypsum (if compacted)	Improves structure and calcium availability; reduces hollow heart or cracking.
At Planting	Basal Dose (NPK + S + Zn + B)	Supports shoot growth, root establishment, and proper stolon/tuber initiation.
At Planting	Proper Spacing (20–25 cm plants, 60 cm rows)	Ensures uniform tuber size, canopy, and light interception.
Vegetative (15–30 DAS)	1st Urea Top-Dress + Irrigation	Supports shoot/leaf growth; moisture critical for stolon formation and uptake.
Vegetative	Weeding + Light Earthing Up	Reduces competition and supports early tuber bulking.
Tuber Initiation (30–45 DAS)	2nd Urea Split + Potash + Micronutrients (B + Zn)	Potash improves tuber size/skin/shelf-life; Boron supports uniform formation.
Tuber Initiation	Maintain Moisture Uniformly	Critical stage—stress or waterlogging reduces tuber number and quality.
Bulking (45–70 DAS)	Optional Urea Split + K Fertilizer	Supports continu afdhjf afjfhja fajfkaf jfa-jfkaj ed growth and sugar accumulation.
Bulking	Avoid Excess Nitrogen	Too much N ' foliage heavy, poor tuber development.
Bulking	Second Earthing Up (optional)	Prevents greening; supports bulking.
Maturation (70–90 DAS)	Stop N + Controlled Irrigation	Lets plants mature and tubers harden; reduces rotting.
Pre-Harvest	Irrigation withdrawal 10–15 days before harvest	Aids skin setting and easier harvest.
Post Harvest	Residue incorporation + Legume green manure	Restores organic matter, structure, and reduces disease carryover.

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